Workshop 2.1: Interoperability Standards & Architecture

Phase 2: Digital Collaboration

Workshop Code: WS 2.1
Primary Component: Interoperability Platform
Duration: 2 days
Timing: Month 4, Week 3-4
Phase Theme: "Connecting Systems, Connecting Nations"

Executive Summary

This workshop shifts from foundational building blocks to actual integration—designing the Government Interoperability Platform that will enable seamless data exchange across Trinidad & Tobago's Ministries, Departments, and Agencies (MDAs). Drawing heavily on Estonia and Finland's X-Road success, participants will define technical standards, API architectures, and governance frameworks to break down government silos. The platform designed here will become the digital "nervous system" connecting e-ID authentication, MDA services, and the data centre infrastructure into a coherent ecosystem.

Key Outcome: Interoperability platform technical architecture and standards framework enabling government-to-government data exchange while maintaining security and privacy.

Workshop Objectives

Primary Objectives

1. **Define Interoperability Architecture:** Establish technical architecture for government interoperability platform
2. **Adopt Standards:** Select international standards for APIs, data exchange, security, and semantic interoperability
3. **X-Road Adaptation:** Evaluate Estonia/Finland X-Road model for T&T implementation
4. **Break Down Silos:** Map current MDA systems and design integration approach
5. **Security & Privacy Framework:** Ensure data exchange maintains confidentiality, integrity, and audit trails

Secondary Objectives

- Build technical capacity on interoperability concepts
- Identify quick-win integration use cases
- Establish API governance model
- Plan for regional expansion (foundation for WS 2.2)
- Integrate GID principles from WS 1.3

Target Participants (35-45 people)

Core Stakeholders (Required)

- **MPAAI Interoperability Team** (5-6)
- Interoperability Platform Director
- Chief Technical Architect
- API governance leads

- Security architects

- **MDA Technical Representatives** (12-15)

- IT directors/CTOs from 8-10 key MDAs:
    - Health, Education, Social Services
    - Revenue Authority, Customs
    - Immigration, Police
    - Business registration
    - Land registry

- At least 1-2 technical staff per MDA

- **National-CSIRT** (3-4)

- Cybersecurity architects
- Security certification specialists
- Incident response team

Extended Stakeholders (Important)

- **Current System Vendors** (4-6)
- Representatives from major government IT vendors
- Technical architects familiar with existing systems

- Integration specialists

- **Data Protection/Legal** (2-3)
- Privacy officers
- Legal advisors on data sharing agreements
- **GID Advisory Committee** (2)
- Representatives from WS 1.3 to ensure inclusion principles

Expert Resources

- **EU Interoperability Experts** (3-4)
- X-Road architect from Estonia or Finland
- EU Interoperability Framework (EIF) specialist
- API governance expert
- Semantic interoperability specialist
- **CAF & EU Delegation** (2)
- **Technical Consultants** (2-3)

Detailed Agenda

DAY 1: Interoperability Fundamentals & Architecture

Morning Session (9:00 AM - 12:30 PM)

9:00 - 9:30 | Opening & Context Setting
- Welcome and workshop objectives
- **The Problem:** Current state of government silos
- Stories of citizens providing same data multiple times
- MDAs unable to verify information
- Manual file transfers and email attachments
- **The Vision:** "Once-only" principle—government knows what it needs to know
- Connection to WS 1.1 (e-ID) and WS 1.2 (Data Centre)

9:30 - 11:00 | Interoperability Fundamentals
- **Presentation:** "What is Interoperability?" (25 min)
- Technical interoperability (systems can exchange data)
- Semantic interoperability (systems understand the data)
- Organizational interoperability (processes and governance)

- Legal interoperability (data sharing agreements, privacy)
- EU Interoperability Framework (EIF) overview

- **Case Study:** "Estonia's X-Road: 20 Years of Success" (35 min)
- Presented by Estonian X-Road expert
- Architecture overview
- Security model (end-to-end encryption, logging)
- Governance (who can access what)
- Real-world use cases (health records, tax filing, business registration)
- Lessons learned and evolution

- Numbers: 52,000+ organizations, 2,800+ services, 1 billion queries/year

- **Interactive Q&A** (30 min)

- Deep dive into X-Road's applicability to T&T

11:00 - 11:15 | BREAK

11:15 - 12:30 | Current State Assessment
- **Facilitated Session:** "Mapping the T&T Government IT Landscape" (75 min)
- Pre-prepared visual map of major government systems (on large wall chart)
- MDA representatives add details:
- What systems do you have?
- What data do they hold?
- Who needs access to your data?
- Whose data do you need?
- Current integration points (if any)
- Technology platforms and vendors

- Facilitator maps connections, identifies:

  - Heavily siloed agencies
  - Common data needs
  - Duplicate data collection
  - Missing connections
  - Technical barriers

- **Key Insights Discussion** (15 min)

- What patterns emerge?
- Where are the biggest pain points?
- Where are quick-win opportunities?

12:30 - 1:30 PM | LUNCH

Afternoon Session (1:30 PM - 5:00 PM)

1:30 - 3:00 | Technical Architecture Deep Dive
- **Presentation:** "Interoperability Platform Architecture Options" (30 min)
- Centralized hub model
- Federated/distributed model (X-Road approach)
- Hybrid approaches
- API Gateway vs. Service Mesh vs. Data Exchange Layer
- Security components (encryption, authentication, authorization, logging)
- Comparison of architectures

- **Technical Workshop:** "Designing T&T Architecture" (60 min)

- Breakout groups (mixed MDA + technical experts):

    - **Group 1:** Core Platform Architecture
    - Centralized vs. distributed
    - Technology stack options

    - Scalability and performance

    - **Group 2:** Security Architecture

    - Authentication (link to e-ID from WS 1.1)
    - Authorization models
    - Encryption (in transit, at rest)

    - Audit logging and compliance

    - **Group 3:** Data Exchange Patterns

    - Synchronous vs. asynchronous
    - Request-response vs. publish-subscribe
    - Real-time vs. batch

    - Data transformation and mapping

    - **Group 4:** Integration with Existing Systems

    - Legacy system integration strategies
    - Adapter/connector patterns
    - Vendor cooperation requirements

Each group produces architecture sketch and key decisions

- **Report Back & Discussion** (30 min)

3:00 - 3:15 | BREAK

3:15 - 5:00 | Standards & Specifications
- **Presentation:** "Interoperability Standards Landscape" (30 min)
- **API Standards:** REST, GraphQL, gRPC comparison
- **Data Formats:** JSON, XML, Protocol Buffers
- **Security:** OAuth 2.0, OpenID Connect, mutual TLS
- **Messaging:** SOAP (legacy), REST APIs (modern)
- **Semantic Standards:** Common data models, taxonomies, ontologies
- **Metadata:** Dublin Core, DCAT for data catalogues
- **Government-specific:** UN/CEFACT, GovStack specifications

- **Interactive Session:** "Standards Selection" (45 min)
- Facilitated discussion to select standards for T&T:
  - Which API style? (Recommendation: RESTful with OpenAPI specs)
  - Which authentication? (Link to e-ID OAuth/OIDC)
  - What data formats?
  - What documentation standards?
- Decision matrix: Factors to consider
  - Maturity and stability
  - Industry adoption
  - Vendor support
  - Learning curve
  - Future-proofing
- **Consensus Building** (15 min)
- Draft "T&T Government Interoperability Standards" document outline

5:00 | Day 1 Close
- Summary of architectural decisions
- Homework: Review use case materials for Day 2

DAY 2: Use Cases, Governance & Implementation

Morning Session (9:00 AM - 12:30 PM)

9:00 - 9:15 | Day 1 Recap & Day 2 Preview

9:15 - 10:45 | Use Case Development
- **Presentation:** "Designing Interoperability Use Cases" (20 min)
- Use case template
- Business value assessment

- Technical complexity assessment
- Priority matrix (value vs. effort)

- **Breakout Workshop:** "Priority Use Cases" (70 min)

- 4-5 groups, each develops 2-3 use cases:

Suggested use cases (groups select and detail):
- **Health:** Patient medical records accessible to authorized providers
- **Social Services:** Eligibility verification for benefits (income, family status)
- **Business:** Single business registration across multiple agencies
- **Tax/Revenue:** Income verification from employers for tax filing
- **Education:** Student records transfer between institutions
- **Immigration:** Background checks across police, courts, tax
- **Land:** Property ownership verification for mortgages, sales
- **Emergency Services:** Integrated emergency response (health, police, fire)

For each use case, specify:
- Business problem solved
- Participating agencies (data providers and consumers)
- Data elements needed
- Security/privacy requirements
- Legal/consent requirements
- User experience improvement
- Technical integration approach
- GID considerations (from WS 1.3)

- **Gallery Walk** (20 min)
- Groups post use cases
- Participants vote on priorities (dot voting)

10:45 - 11:00 | BREAK

11:00 - 12:30 | API Design Workshop
- **Presentation:** "RESTful API Design Best Practices" (25 min)
- Resource-based URLs
- HTTP methods (GET, POST, PUT, DELETE)
- Status codes and error handling
- Versioning strategies
- Pagination for large datasets
- Rate limiting
- OpenAPI/Swagger documentation

- **Hands-On Exercise:** "Design an API" (55 min)
- Groups take one of their priority use cases

- Design the API specification:
    - Endpoints and resources
    - Request/response formats (JSON schemas)
    - Authentication and authorization
    - Error scenarios
    - Documentation
- Template provided (OpenAPI format)
- **Share & Critique** (10 min)
- One group presents, others provide feedback

12:30 - 1:30 PM | LUNCH

Afternoon Session (1:30 PM - 5:00 PM)

1:30 - 2:45 | Data Governance & Privacy
- **Presentation:** "Data Governance for Interoperability" (25 min)
- Data ownership vs. data stewardship
- Data sharing agreements (legal framework)
- Consent management
- Purpose limitation (data used only for stated purpose)
- Retention and deletion policies
- Audit trails and accountability
- GDPR principles applied (from WS 1.3)

- **Panel Discussion:** "Legal and Privacy Framework" (30 min)
- Legal experts
- Data protection authority
- MDA representatives
- Topics:
    - What legislation enables/restricts data sharing?
    - What consent is required from citizens?
    - How to handle sensitive data (health, criminal records)?
    - Cross-border data sharing considerations (preview WS 2.2)
- **Interactive Exercise:** "Data Sharing Agreement Template" (20 min)
- Review template for bilateral MDA data sharing
- Identify key clauses and requirements

2:45 - 3:00 | BREAK

3:00 - 4:15 | Governance Model
- **Presentation:** "API Governance Framework" (20 min)
- Who approves new APIs?
- How are APIs registered and discovered? (API catalogue)
- Who monitors usage and performance?
- How are changes managed (versioning, deprecation)?
- Quality assurance and testing requirements
- Documentation standards
- Service level agreements (SLAs)

- **Workshop:** "T&T Interoperability Governance" (45 min)

- Design governance structure:

  - **Governance Board** (decision-making, policy)
  - Membership
  - Meeting frequency
  - Authority
  - **Technical Committee** (standards, architecture)
  - **API Catalogue** (registration, discovery)
  - **Change Management Process**
  - **Dispute Resolution**
  - **Compliance and Auditing**

- Draft governance charter outline

4:15 - 5:00 | Implementation Roadmap & Next Steps
- **Presentation:** "From Design to Deployment" (15 min)
- Phased implementation approach
- Pilot phase (2-3 use cases, limited MDAs)
- Expansion phase (more use cases, more MDAs)
- Maturity phase (full ecosystem)
- Link to regional expansion (WS 2.2)

- **Collaborative Roadmap Planning** (20 min)
- Timeline for next 12 months:

  - Month 5-6: Platform development begins, pilot use case selection
  - Month 7-9: Pilot implementation and testing
  - Month 10-12: Initial production deployment, monitoring

- Dependencies and risks

- **Action Items & Commitments** (10 min)

- Working groups formation (technical, governance, legal)
- Follow-up tasks with owners

- Preparation for WS 2.2 (Regional Cooperation)
- **Workshop Evaluation & Closing Remarks** (15 min)

Key Discussion Questions

Architecture Questions

- Should T&T adopt X-Road directly, adapt it, or build custom?
- What level of centralization is appropriate for T&T context?
- How do we handle legacy systems that can't easily integrate?
- Where should the platform be hosted (data centre from WS 1.2)?

Standards Questions

- Which API style best fits government needs (REST, GraphQL, etc.)?
- Should we mandate standards or allow flexibility?
- How do we ensure vendor compliance with standards?
- What documentation requirements are realistic?

Governance Questions

- Who has authority to approve data sharing between MDAs?
- How do we balance agency autonomy with centralized governance?
- What penalties for non-compliance with standards?
- How do we fund and sustain the platform?

Privacy & Security Questions

- How do we prevent unauthorized access to sensitive data?
- What audit trails are needed for accountability?
- How granular should access control be (agency-level, user-level, purpose-level)?
- How do we handle citizen consent for data sharing?

Expected Outputs & Deliverables

Immediate Outputs (During Workshop)

1. **Interoperability Platform Architecture** (draft)
2. Core components and technology choices
3. Security model

4. Integration patterns
5. **Technical Standards Framework**
6. API standards (REST, OpenAPI, OAuth 2.0, etc.)
7. Data format standards
8. Documentation requirements
9. **Priority Use Cases** (10-15 detailed)
10. Ranked by value and feasibility
11. Technical specifications for top 3-5
12. **API Specification Examples** (2-3 sample APIs designed)
13. **Governance Framework** (draft)
14. Governance board structure
15. Technical committee
16. Decision-making processes
17. **Data Sharing Agreement Template**
18. **Implementation Roadmap** (12-month view)

Post-Workshop Deliverables (2 weeks)

1. **Workshop Report** (Product 3)
2. Executive summary
3. Architecture decisions and rationale
4. Standards selection
5. Use case catalogue
6. Governance framework
7. Implementation roadmap
8. Next steps
9. **T&T Government Interoperability Standards** (Appendix)
10. Comprehensive standards document
11. Technical specifications
12. Compliance requirements
13. **API Design Guidelines** (Appendix)
    - Developer-friendly guide

    - Code examples
    - Best practices

14. **Governance Charter** (Appendix)
    - Detailed governance model
    - Roles and responsibilities
    - Processes and procedures

Success Metrics

Participation Metrics

- 80%+ of targeted MDAs represented
- Active participation from National-CSIRT
- EU expert engagement throughout

Engagement Metrics

- 10+ priority use cases identified
- Consensus on architecture approach
- Standards framework adopted
- Governance model agreed

Satisfaction Metrics

- 4.2/5 average participant satisfaction
- 85%+ understand interoperability concepts
- 90%+ see value for their agency

Outcome Metrics

- Technical architecture ready for procurement
- At least 3 pilot use cases selected
- Governance board members identified
- Timeline for implementation established

Connection to Other Workshops

Builds On:

- **Workshop 1.1 (e-ID):** e-ID provides authentication layer for interoperability platform
- **Workshop 1.2 (Data Centre):** Platform will be hosted on infrastructure designed in WS 1.2
- **Workshop 1.3 (GID):** Accessibility, privacy, and inclusion principles embedded in architecture

Builds Foundation For:

- **Workshop 2.2 (Regional Cooperation):** Architecture designed here extends to regional partners
- **Workshop 2.3 (Cross-Border e-ID):** Interoperability platform enables cross-border services
- **Workshop 3.4 & 4.1 (MDA Showcases):** Use cases developed here will be demonstrated
- **Workshop 4.2 (Governance):** Governance model formalized

Inputs From:

- **Product 2:** Technical Progress Analysis 1 (Q1 progress report)
- **e-ID architecture** from WS 1.1 follow-up work

Feeds Into:

- **Platform procurement** (Month 5-6)
- **Pilot implementation** (Month 7-9)
- **Product 4:** Partial Implementation Report (Q2)

Pre-Workshop Preparation

6 Weeks Before

- [ ] Confirm date and book venue
- [ ] Send save-the-date to all targeted MDAs
- [ ] Book EU X-Road and interoperability experts
- [ ] Begin current state assessment (pre-work)

4 Weeks Before

- [ ] Conduct pre-workshop interviews with MDAs (understand their systems)
- [ ] Prepare current state map (visual of government IT landscape)

- [ ] Finalize detailed agenda
- [ ] Send formal invitations

2 Weeks Before

- [ ] Send pre-reading materials:
- X-Road case study
- EU Interoperability Framework overview
- RESTful API design basics
- Current state assessment results
- [ ] Request MDAs bring technical documentation of their systems
- [ ] Finalize catering and logistics

1 Week Before

- [ ] Send reminder with logistics and preparation requests
- [ ] Print large-format current state map for annotation
- [ ] Prepare use case templates and worksheets
- [ ] Technical rehearsal for demos

Materials & Resources Needed

Presentation Materials

- PowerPoint slides for all sessions
- X-Road demo/walkthrough (video or live)
- Architecture diagrams and examples
- API specification examples

Participant Materials

- Workshop agenda and participant guide
- Current state map (large poster for annotation)
- Use case template worksheets
- API design worksheets (OpenAPI templates)
- Technical standards reference guide
- Governance framework template
- Notebook and pen

Technology

- Projector and screen
- Laptops for technical exercises (API design)

- Demo environment (if X-Road demo is live)
- WiFi access
- Flip charts and markers
- Large wall space for mapping exercise

Documentation

- Attendance tracking
- Photography for documentation
- Note-takers for each breakout session
- Technical note-taker for architecture discussions

Budget Estimate

Category	Item	Estimated Cost
Venue	2-day conference room + 4-5 breakout rooms	$3,500
Catering	Coffee breaks (4), lunches (2)	$4,500
Travel	EU experts flights & accommodation (3-4 experts)	$22,000
Materials	Printing (large maps, worksheets), supplies	$1,000
Technology	AV equipment, laptops for exercises	$1,500
Demo Environment	Cloud hosting for X-Road demo (if needed)	$500
Facilitation	Professional facilitator with interoperability expertise	$5,000
Contingency	10%	$3,800
TOTAL		**$41,800**

Risk Mitigation

Risk	Mitigation
MDAs resistant to sharing data	Emphasize once-only principle benefits; address privacy concerns; show EU examples; pilot with willing agencies first
Technical complexity overwhelming	Balance technical depth with accessible explanations; provide glossary; pair technical and policy staff
Unrealistic expectations (wanting everything immediately)	Phased roadmap; clear prioritization; pilot approach; manage expectations
Vendor lock-in concerns	Emphasize open standards; evaluate open-source options (X-Road is open source); multiple vendor support
Lack of consensus on architecture	Present as "recommendation" not final decision; allow refinement during implementation; expert guidance
Security concerns paralyze progress	Strong security framework from start; National-CSIRT involvement; EU security model examples

Gender & Inclusion Considerations

Representation

- Ensure women technical leads invited from MDAs
- GID Advisory Committee representatives present
- Diverse technical expertise

Content

- Use cases include services for vulnerable populations (from WS 1.3)
- Accessibility requirements for APIs (screen readers can access services)
- Privacy protections especially for sensitive populations (GBV survivors, etc.)
- Multi-channel service delivery (not digital-only)

Process

- Technical jargon explained

- Multiple participation modes
- Value both technical and policy/user perspectives

Follow-up Actions (Post-Workshop)

Week 1

- [ ] Send thank you emails to all participants
- [ ] Establish technical working group (first meeting within 1 week)
- [ ] Begin drafting workshop report
- [ ] Share architecture diagrams and decisions with procurement team

Week 2

- [ ] Finalize and circulate draft workshop report
- [ ] Publish draft standards document for MDA review
- [ ] Schedule governance board inaugural meeting
- [ ] Begin pilot use case selection process

Week 3-4

- [ ] Incorporate feedback and publish final report
- [ ] Submit to CAF and EU as Product 3
- [ ] Finalize standards document
- [ ] Begin platform procurement planning

Month 5-6

- [ ] Platform vendor selection and contracting
- [ ] Pilot use case detailed design
- [ ] Prepare for Workshop 2.2 (Regional Cooperation—extend architecture regionally)
- [ ] MDA onboarding and capacity building

Links to TOR Requirements

This workshop directly addresses:

TOR Section 5, Objective B: Interoperability Platform Actions
- **Action a:** "Coordinate and host regional workshops with 2 to 3 Caribbean countries to develop a government-to-government interoperability platform." [This is the national foundation; WS 2.2 extends regionally]

- **Action b:** "Support the development of software components and API integrations for the regional interoperability network ensuring the public officer's systems work together without manual hand-offs or data silos."
- **Action d:** "Ensure that all interoperability activities are carried out in adherence to the best international practices, standards, and digital solutions, to maximize security, accessibility, and efficiency."

TOR Section 6, Services:
- **Service 2:** "Support oversight and monitoring of all project components."
- **Service 3:** "Support effective contract management, document control, and tracking of contract milestones."
- **Service 6:** "Actively seek and evaluate international solutions, technologies, and best practices..." [X-Road evaluation]

TOR Section 7, Products:
- **Product 3:** Workshop Reports

Document Status: Draft for Review
Version: 1.0
Date: September 26, 2026
Next Review: 2 weeks before workshop execution

TECHNICAL NOTE: This workshop is highly technical. Ensure facilitators can bridge technical and policy audiences. Consider pre-workshop technical primer for non-technical participants.